# A Predictive Framework for Optimizing Resource Allocation in Concurrent Software Development

## *Introduction-*

**Software projects are usually complicated endeavors that include hundreds of various tasks. The primary challenge during their development consists in the allocation of the limited number of resources among different types of work. In such a case, it becomes particularly important to ensure that the system's functioning is optimized by following specific rules.**

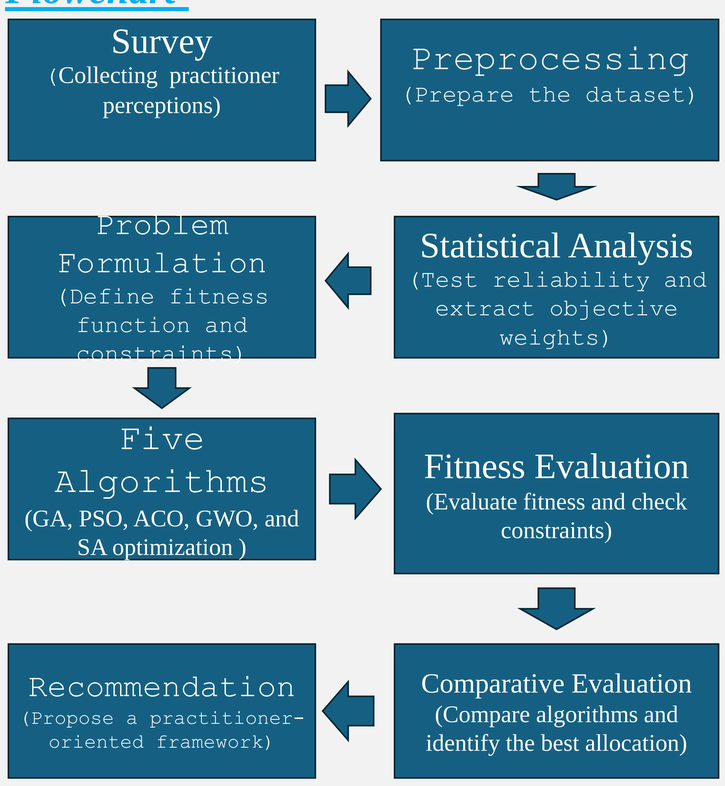
## *Research Design (3 Phases)-*

***Phase 1 –* Quantitative Survey: A cross-sectional online survey of 47 Bangladeshi software professionals to assess resource-allocation challenges and objective priorities.**

***Phase 2 –* Metaheuristic Optimization: Survey-derived weights are used to formulate a multi-objective allocation problem solved by GA, PSO, ACO, GWO, and SA.**

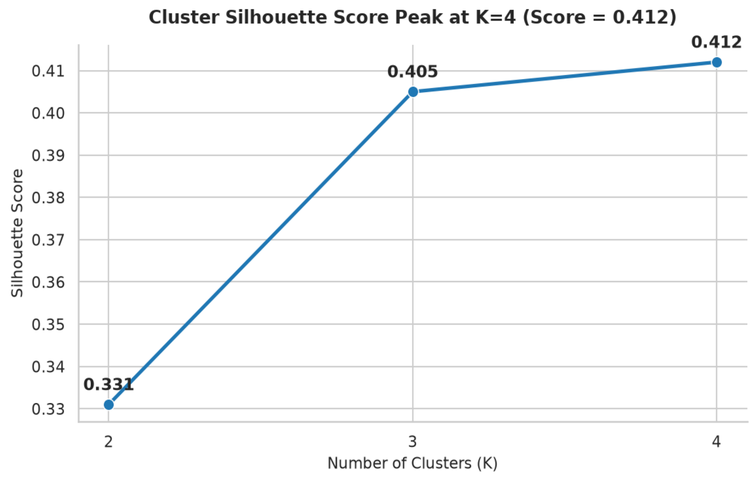
***Phase 3 –* Comparative Evaluation: The five algorithms are compared based on solution quality, convergence, and constraint satisfaction.**

## *Flowchart-*

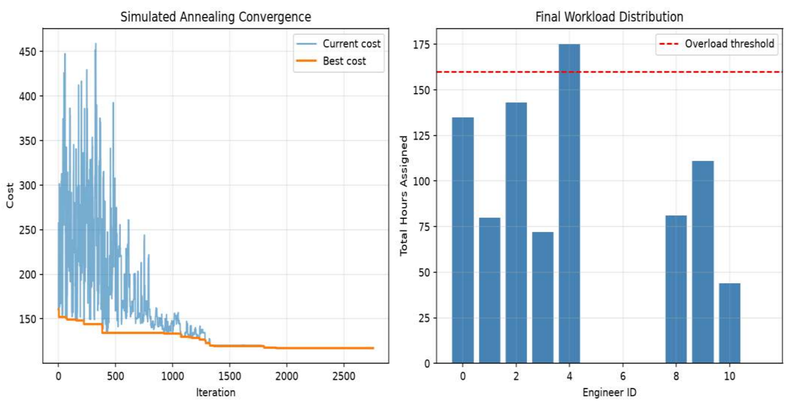



## *Data Analysis-*

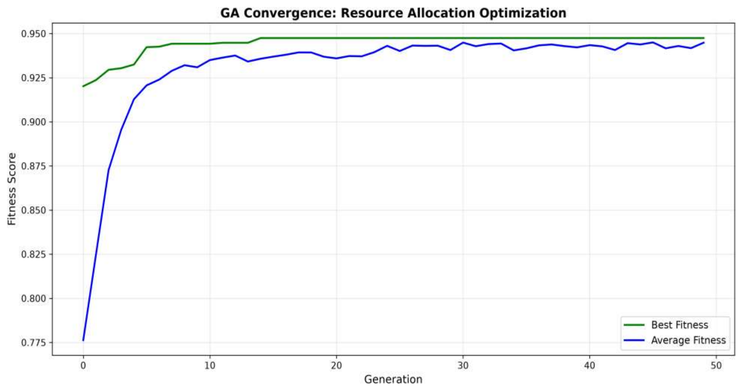
The data analysis proved that the survey was highly reliable and produced optimisation insights that could be put to use. The generalisability is limited by the small sample size, single country scope, synthetic instances, and scalarization.




### Silhouette Score




### Sa convergence




### Ga convergence

## *Results and Discussion*

The survey demonstrated high internal consistency (Cronbach's α = **0.865–0.939**). Objectives (M = **4.036**) and Evaluation (M = **4.027**) scored higher than Challenges (M = **3.414**), with skill mismatch as the leading challenge. No significant differences occurred across roles, experience, or organization sizes (**p > 0.05**).

K-Means identified four profiles, led by Acceptance-Dominant (**46.8%**). GA improved fitness by **2.97%** with perfectly balanced workloads (SD = 0.00), while GWO achieved the lowest cost (**≈117**, vs. ACO ≈120 and SA ≈125) on the shared 12 × 8 instance. All three exceeded capacity by reaching **175h vs. 160h**. Findings support objective-driven algorithm selection; however, **n = 47** and differing test scales limit generalizability and direct comparison.

| *Key* | *Analysis Area* | *Result* | *Interpretation* |
|---|---|---|---|
| Cronbach's Alpha | Survey Reliability | RC: 0.939<br>AC: 0.893<br>IA: 0.865 | Good internal correlation |
| Mean Score | Practitioner perception | SD=1.2<br>Mean=3.767 | Strong favorable attribute |
| P-value | Group difference | P>0.05 | No difference between role , experience and org size |
| K-means clustering | Segmentation | K=4<br>Silhouette=0.412 | 4 distinct profiles |
| Fitness and workload | Genetic algorithm | SD=0.00<br>Fitness=0.9475 | Perfectly balanced workload |
| Cost and workload | GWO/ACO/SA | 117,120,125 | Shows lower cost |

## *Conclusion-*

Aim: To develop a multi-criteria and data driven framework for resource allocation in parallel software development. Methodology : We propose two-step methodology. Firstly, we conduct a survey of 47 software professionals from Bangladesh and then perform a comparative study among five different meta-heuristic algorithms (GA, PSO, ACO, GWO and SA) based on weights obtained from the survey. Contribution: A novel resource allocation framework which uses survey-based weights and comparative study of meta-heuristic algorithms along with K-Means segmentation is presented.


### Acknowledgment-

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